De-identified Clinical Case: Severe / Difficult-to-Treat Rheumatoid Arthritis

Structured research brief prepared as input for an automated dataset & literature discovery harness

De-identification notice. This document contains **no PII/PHI** by design: no names, dates of birth, calendar dates, geographic locations, institution names, record numbers, or other identifiers. All timing is expressed as relative durations. Clinician roles are given by specialty only. Drug and disease terms are generic. A synthetic case label is used purely as a handle.

Purpose & limits. This is a research-and-hypothesis-generation aid, **not medical advice, diagnosis, or a treatment recommendation.** Every direction below is a candidate to investigate and to discuss with the licensed care team. Items not established in the source record are marked **TO CONFIRM** and were not invented.

1 Case snapshot

Case label	CASE-RA-D2T-A (synthetic; not an identifier)
Subject	Adult female
Illness duration	Approximately one decade; multi-system, chronic
Primary analytic focus	Severe, long-standing, apparently treatment-refractory rheumatoid arthritis — candidate **EULAR ‘difficult-to-treat RA’ (D2T RA)**
Broader phenotype	Overlapping autoimmune picture: inflammatory arthritis + gastrointestinal / IBD involvement + autoimmune history, with an active neurological work-up in progress
Most urgent flagged issue	A ~6-month gap in dedicated disease-modifying arthritis therapy
Care setting	Records span three separate health systems (specialty referrals only, de-identified)
Analytic goal	Surface data-driven, evidence-graded hypotheses for achieving RA remission, including non-obvious / precision / horizon strategies

2 Multi-system problem list (de-identified)

System	Status	Notes for analysis
Inflammatory arthritis / RA	Active, severe, ~decade	Incompletely controlled; recent ~6-month gap in DMARD-directed therapy. Primary focus.
Additional autoimmunity	Followed by rheumatology	Autoimmune history; overlap features to be characterized (serologies TO CONFIRM).
Gastrointestinal / IBD	Followed by GI / IBD service	Relevant: several agents are cross-indicated for IBD + inflammatory arthritis.
Cardiac	Work-up unresolved (open)	Contextual comorbidity; relevant to drug-class safety selection (e.g., JAKi CV signals).
Neurological (bulbar)	Active work-up	New voice / swallowing / breathing symptoms; see contingency note below.

System	Status	Notes for analysis
Sleep	Study never completed (open)	Open item; may confound fatigue / symptom burden.
Surgical / device	Implant status unconfirmed (open)	Open item to resolve in record.

3 Rheumatologic disease detail (primary focus)

The core analytic problem is long-standing, severe RA that has not reached sustained remission over roughly a decade, now compounded by a multi-month lapse in disease-modifying therapy. This profile is what the EULAR framework labels **difficult-to-treat RA** — typically defined by failure of multiple mechanism-of-action classes together with signs or symptoms suggesting active or progressive disease. Approximately 5–20% of RA patients are refractory to all available biologic classes, so a structured, data-driven approach to endotyping and next-line selection is warranted rather than simply cycling drugs empirically.

Established from record

- Anti-TNF biologic exposure is documented (an anti-TNF agent, certolizumab pegol, appears in the record).
- Disease is chronic (~decade) and severe, with rheumatology actively involved.
- A ~6-month gap in dedicated arthritis-directed treatment is the highest-priority actionable issue identified.

TO CONFIRM from source records (high-value unknowns for any modeling)

- **Serostatus:** rheumatoid factor and anti-CCP / ACPA (positive vs negative, titers).
- **Disease-activity trajectory:** DAS28-CRP / DAS28-ESR, CDAI, SDAI; swollen and tender joint counts over time.
- **Acute-phase reactants:** CRP and ESR trend.
- **Structural damage:** erosive vs non-erosive; radiographic progression.
- **Full therapy sequence:** every csDMARD (e.g., methotrexate, leflunomide, sulfasalazine, hydroxychloroquine), tsDMARD (JAK inhibitor), and biologic (IL-6R, B-cell / anti-CD20, T-cell co-stimulation) — with doses, durations, response, adverse events, and reason for each discontinuation.
- **Failure mechanism:** anti-drug-antibody / drug-level testing where a biologic ‘failed’ (distinguishes pharmacokinetic failure from true mechanistic non-response).
- **Extra-articular features:** nodules, interstitial lung disease, vasculitis, secondary Sjögren, etc.
- **Overlap serologies & genetics:** ANA / ENA panel; HLA data if available.
- **Confounders of apparent refractoriness:** fibromyalgia / central sensitization assessment; adherence history.

4 Overlap context & neurological contingency

The concurrent GI / IBD involvement and broader autoimmune history matter analytically: they raise the value of agents with cross-indication efficacy and of searching for a single upstream immune axis that could address multiple systems at once, rather than treating each in isolation.

An active neurological work-up addresses new bulbar symptoms (voice, swallowing, breathing). A brain MRI was verbally summarized as normal, but the written report notes a prominent posterior-circulation vessel in contact with the dorsolateral medulla — a possible neurovascular-contact finding in the appropriate clinical context, documented as a contingency pathway. A motor-neuron-disease consideration sits on the differential (with nerve conduction / EMG as the key discriminating test), and antibody testing for a neuromuscular-junction disorder has been flagged as a straightforward next step given the symptom cluster plus autoimmune background. This neurological thread is contextual: the harness's primary target is RA remission, but the overlap is analytically relevant and should not be discarded.

Note: the neurological items are provided as context only. They are an active clinical work-up owned by the care team and are not the object of the data-science search.

5 Structured feature set (for computational matching)

Machine-readable equivalents ship in the companion JSON. 'Unknown' entries are themselves signal — they tell the loop what to go find.

Feature	Value / status
primary_condition	rheumatoid_arthritis (severe, chronic)
refractory_class	candidate difficult-to-treat RA (EULAR D2T)
disease_duration	~10 years
sex	female
seropositivity (RF, anti-CCP)	UNKNOWN — to confirm
disease_activity_scores	UNKNOWN — to confirm (DAS28 / CDAI / SDAI)
erosive_disease	UNKNOWN — to confirm
csDMARD_history	UNKNOWN / partial — to confirm
anti_TNF_exposure	YES (certolizumab pegol documented)
IL6R / JAKi / anti-CD20 / abatacept history	UNKNOWN — to confirm
anti_drug_antibody_testing	UNKNOWN — to confirm
current_DMARD_therapy	GAP (~6 months without dedicated arthritis therapy)
comorbid_autoimmune	YES (autoimmune history; GI/IBD involvement)
extra_articular_manifestations	UNKNOWN — to confirm
central_sensitization / fibromyalgia overlap	UNKNOWN — assess (confounds refractoriness)
genetic_HLA_data	UNKNOWN — to confirm

6 Candidate public data resources for the harness to query

Resource	What it offers	How the loop uses it
NCBI GEO	RA synovial & whole-blood transcriptomes (e.g. GSE89408, GSE77298, GSE55235, GSE55457, GSE12021, GSE45867 treatment-response)	Differential expression, endotype signatures, treatment-response markers (GEO2R / limma).
ImmPort	Curated immunology studies incl. synovial datasets (e.g. SDY1599)	Cell-type and pathway signals; cross-study meta-analysis.
AMP RA/SLE Network	Single-cell synovial atlas of RA tissue	Cell-state endotyping (fibroblast / B-cell / macrophage subsets).
PEAC cohort	Synovial molecular pathology in early arthritis	Pathotype (lympho-myeloid / diffuse / fibroid) mapping to response.
R4RA trial data	Biopsy-stratified rituximab vs tocilizumab response + ML models	Endotype-to-drug matching; multidrug-resistance signature.
ArrayExpress / BioStudies	Additional expression & multi-omics	Broaden meta-analysis; validation cohorts.
ClinicalTrials.gov	Active trials in refractory / D2T RA & autoimmune cell therapy	Horizon options, eligibility scan, mechanism trends.
Open Targets / DGIdb / DrugBank	Target–drug–disease associations	Drug-repurposing candidates against implicated targets.
Tools	CIBERSORTx, GSEA, WGCNA	Immune deconvolution, pathway enrichment, module discovery.

7 Seed research directions (‘outside the box’, evidence-graded)

Each is a hypothesis to investigate with data and to review with the care team — not a recommendation. Grades: **[E]** established framework, **[M]** emerging / trial-supported, **[I]** investigational / early-phase.

[E] Reframe formally as EULAR difficult-to-treat RA and run its checklist

First verify the diagnosis (rule out seronegative mimics and overlap connective-tissue disease), then separate inflammatory from non-inflammatory drivers. Fibromyalgia / central sensitization can masquerade as treatment resistance; distinguishing them changes the entire strategy. This is the cheapest, highest-yield first move.

[E] Mechanism-of-action cycling with objective failure criteria

If exposure has been dominated by the anti-TNF class, a structured switch across distinct mechanisms (IL-6R blockade, JAK inhibition, T-cell co-stimulation blockade, B-cell depletion) is standard. Crucially, check anti-drug antibodies and drug levels on prior ‘failures’ to separate pharmacokinetic failure (dose / immunogenicity) from genuine mechanistic non-response — they imply opposite next steps.

[M] Precision / biopsy-guided endotyping

In the R4RA trial, synovial B-cell-poor patients responded far better to IL-6R blockade than to anti-CD20 therapy, and a stromal / fibroblast signature marked patients refractory to all drugs; machine-learning models predicted response and even multidrug resistance. Data-science analog for this case: attempt to infer endotype from available blood transcriptomic signatures and, if clinically appropriate, discuss whether a synovial biopsy could de-risk the next drug choice instead of guessing.

[M] Shared-pathway strategy across the overlap phenotype

With RA + IBD + broader autoimmunity co-occurring, prioritize agents with cross-indication efficacy and hunt for a single upstream axis (B-cell, IL-6 / JAK-STAT) whose modulation could quiet multiple systems at once. A network /

pathway-enrichment analysis over the public datasets can rank shared druggable nodes.

[I] Cell therapy for refractory autoimmune disease (horizon watch)

CD19 CAR-T (e.g., the KYV-101 COMPARE trial in difficult-to-treat RA), dual CD19/BCMA constructs, and CAR-Treg approaches have driven deep B-cell reset and even drug-free remission in refractory autoimmune disease. RA experience is still very early (roughly ten treated patients) and available only in trials, so this is a horizon option to track on ClinicalTrials.gov and raise with a specialist — not a near-term plan.

8 What a good harness run should produce

- A ranked list of candidate remission strategies, each with mechanism, supporting datasets / papers (by ID), evidence grade, feasibility, and the specific data that would confirm or refute it.
- An explicit ‘data to obtain’ list that would most sharpen the analysis (maps to the TO CONFIRM items).
- Endotype hypotheses derived from public transcriptomic data and how they map to specific drug classes.
- A shortlist of active, relevant trials with plain-language eligibility notes.
- A clinician-discussion brief: what to raise with rheumatology first, and why.

Standing disclaimer. De-identified research aid only. Not medical advice, diagnosis, or treatment. All hypotheses require review and decision by the licensed care team. No content here should be acted on clinically without that review.